Project 3 - Complete Observability System

Unified metrics, centralized logs, and distributed traces

1. Introduction

Reliable application operations depend on visibility into performance, errors, and request execution. This project builds a local observability stack for a containerized Node.js service.

2. Abstract

The system combines Prometheus metrics, Grafana dashboards, Loki centralized logs, Grafana Alloy log collection, OpenTelemetry instrumentation, and Jaeger trace visualization. Docker Compose provides a repeatable multi-service environment that supports investigation of application health and request behavior.

3. Tools Used

Technology	Purpose
Node.js + Docker	Instrumented application and containerized deployment
Prometheus	Collection of application and runtime metrics
Grafana	Provisioned dashboard for operational visualization
Loki + Grafana Alloy	Centralized collection and querying of structured application logs
OpenTelemetry Collector	Telemetry processing and trace export
Jaeger	Distributed trace search and visualization

4. Steps Involved in Building the Project

1. Developed a Node.js application with request metrics, runtime instrumentation, structured JSON logging, and trace instrumentation.

2. Configured Prometheus scraping and exposed HTTP request counts, duration histograms, and Node.js metrics.

3. Provisioned Grafana datasources and a six-panel monitoring dashboard from configuration files.

4. Configured Grafana Alloy to collect application logs and forward them to Loki for querying.

5. Connected OpenTelemetry tracing through a collector and visualized traces in Jaeger.

6. Orchestrated and validated the seven-service environment using Docker Compose.

5. Testing and Results

Confirmed that seven containers ran successfully. Prometheus collected application metrics; Grafana displayed dashboard panels; Loki showed centralized JSON logs; and Jaeger displayed traces. Evidence screenshots for the dashboard, log queries, and trace view are available in the project repository.

6. Conclusion

The project demonstrates the three core observability signals - metrics, logs, and traces - in one reproducible development environment. It provides hands-on experience with instrumentation, telemetry pipelines, dashboard provisioning, and troubleshooting workflows.